# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1703**

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**COURSE OUTCOME COVERED**

**CO5:** *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 5**

**Duration : 1 Hour**
**Total Marks:50**

Annexure A

## Technical Presentation

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***Code of Conduct:***

*I **Dhanajeyan J (192324211)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

# Technical Presentation

1. **Reactive Agent-Based Traffic Control System**

   Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. **Goal-Based Agent for Healthcare Diagnosis**

   Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. **Utility-Based Agent for E-Commerce Recommendations**

   Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. **Learning Agent for Personalized Education System**

   Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

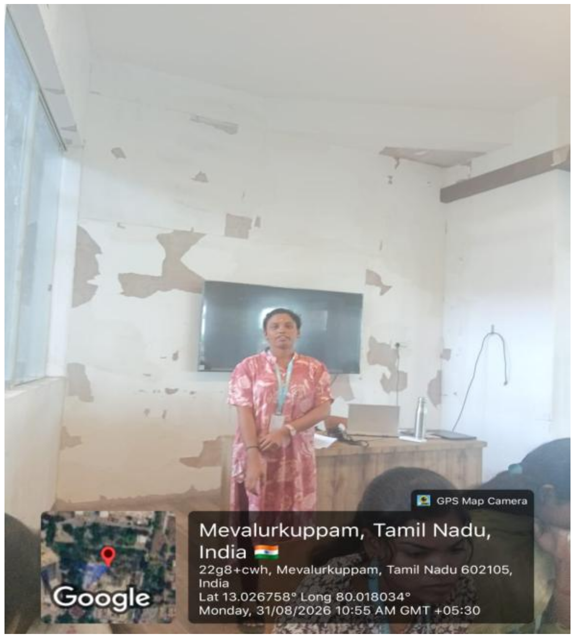
5. **Multi-Agent System for Disaster Management**

   Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Technical Content Accuracy | 30 | Highly accurate, in-depth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| Problem Identification | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| Use of Tools | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| Communication | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| Professionalism | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

**Reactive Agent-Based Traffic Control System**




**1. Introduction**

**A Reactive Agent-Based Traffic Control System is an intelligent traffic management system that uses agents to observe traffic conditions and immediately take suitable actions. Unlike traditional traffic signals that operate using fixed timings, reactive agents continuously monitor the environment and respond to changes such as traffic congestion, accidents, emergency vehicles, and pedestrian movement.**

**The main objective is to reduce traffic congestion, minimize waiting time, improve road safety, and provide efficient traffic flow.**

**2. What is a Reactive Agent?**

**A reactive agent is an intelligent agent that:**

- **Observes the current environment.**
- **Receives information through sensors.**
- **Applies predefined rules.**
- **Takes an immediate action.**
- **Does not necessarily maintain a detailed history of previous situations.**

**For example:**

**IF traffic density is high on Road A THEN increase the green-light duration for Road A.**

**3. System Components**

**Sensors**

- **CCTV cameras**
- **Vehicle detectors**
- **Infrared sensors**
- **GPS information**
- **Emergency vehicle detection systems**

**Agent**

- **Receives traffic information.**
- **Determines the current traffic condition.**
- **Applies decision rules.**
- **Sends commands to traffic signals.**

**Traffic Signal Controller**

- **Controls red, yellow, and green signals.**
- **Changes signal timing according to the agent's decision.**

**Communication Network**

- **Transfers information between intersections and traffic-control agents.**

**ANSWER:**

**1. Introduction**

- **System Definition:** An intelligent traffic management system using autonomous agents to observe real-time conditions and execute immediate actions.
- **Traditional vs. Reactive Systems:**
  - **Traditional Signals:** Rely on fixed, static timing schedules regardless of traffic flow.
  - **Reactive Agents:** Continuously monitor environments to adapt to congestion, accidents, emergency vehicles, and pedestrian flow.

**Key Objectives:**

- Reduce overall traffic congestion.
- Minimize vehicle waiting times.
- Improve road safety and emergency access.
- Provide smooth and efficient traffic flow.

**2. What is a Reactive Agent?** An intelligent decision engine designed to act on immediate environmental inputs.

- **Core Behaviors:**
  - Observes the current state of the environment.

    - Receives operational data directly through sensors.
    - Applies predefined condition-action rules.
    - Takes immediate actions without reliance on long historical data.
- **Condition-Action Rule Example:**
    - **IF** traffic density is high on Road A
    - **THEN** increase the green-light duration for Road A.

### 3. System Components Breakdown

- **Sensors (Input Layer):**
    - CCTV cameras & Vehicle detectors
    - Infrared sensors & GPS data streaming
    - Emergency vehicle detection systems
- **Agent (Decision Layer):**
    - Ingests real-time traffic data.
    - Evaluates immediate road conditions.
    - Executes decision logic and dispatches control commands.
- **Traffic Signal Controller & Communication Network (Execution Layer):**
    - Controls physical red, yellow, and green signals based on agent commands.
    - Transfers real-time data seamlessly between intersections and agents.

### Sensors

- **CCTV cameras**
- **Vehicle detectors**
- **Infrared sensors**
- **GPS information**
- **Emergency vehicle detection systems**

### Advantages

- **Real-Time Responsiveness:** Continuously monitors the environment and takes immediate actions to adapt to dynamic traffic changes.
- **Reduced Congestion & Delay:** Directly works to minimize vehicle waiting times, relieve congestion, and ensure smoother traffic flow compared to static timing signals.
- **Priority Handling:** Instantly responds to critical road events such as emergency vehicles, accidents, and pedestrian movement.
- **Low Computational Complexity:** Because the agent acts on predefined condition-action rules without storing or analyzing detailed past histories, decision-making happens with minimal computational delay.
- **Enhanced Safety:** Dynamically adjusts signal timing to lower collision risks and improve overall road safety.

### Disadvantages

- **Lack of Long-Term Memory:** Does not maintain a detailed history of previous situations, meaning it cannot learn from past trends or predict future traffic patterns.
- **Single Intersection Focus (Suboptimal Global Control):** Making localized, instant decisions for one signal might accidentally cause unexpected traffic bottlenecks at neighboring intersections.

- **Heavy Reliance on Hardware:** Requires continuous data from sensors (CCTV, infrared, GPS, vehicle detectors); sensor failure directly impacts the agent's ability to apply decision rules accurately.
- **Rigid Rule Set:** Operates strictly on fixed $IF \text{ condition } THEN \text{ action}$ rules, making it less adaptable to highly complex or unprecedented traffic scenarios without manual updates.